

IRTc: Copula-Based Item Response Theory

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2026-06-30

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Chapter 1

IRTc: Copula-Based Item Response Theory

Chapter 2

About this thesis

This is the landing page for Damian Betebenner’s dissertation, “IRTc: Copula-Based Item Response Theory”, at dataimago (Independent research — educational measurement & psychometrics (dataimago)).

The thesis is rendered via Quarto + XeLaTeX to a PDF whenever you push changes to `ui/www/` or `R/`. See `docs/thesis.pdf` for the latest build (a stable copy is committed by the `build-thesis.yml` workflow).

2.1 Chapters

- [Front matter](#)
- [Introduction](#)
- [Literature review](#)
- [Methodology](#)
- [Results](#)
- [Discussion](#)
- [Conclusion](#)
- [Back matter](#)

2.2 Formatting

See [THESIS-CLS-README.md](#) for the 3-mode thesis class file strategy + how to iterate on your `thesis.cls` for institutional compliance.

Chapter 3

Front matter

Abstract

Your dissertation abstract — typically one paragraph, ~250–350 words depending on institutional convention. Pull from your `dataimago-spec.yaml`'s `vertical.dissertation.thesis.contribution` field as a starting point.

Acknowledgments

Acknowledgments for your advisor, committee, family, funders. If you have funding to acknowledge, also fill in your `dataimago-spec.yaml`'s `funding` section — that field can be used to auto-render this section in a future iteration.

Dedication

Optional. Remove this section if not used.

Chapter 4

Introduction

4.1 Background

Pull from your `dataimago-spec.yaml`'s `vertical.dissertation.thesis.background` for intellectual + personal context. Two or three paragraphs covering both the intellectual genealogy (what work led you here, what scholarly conversations you're joining) and the personal / practical motivation (why this question, why now, why you).

4.2 Research question

Pull from your `dataimago-spec.yaml`'s `vertical.dissertation.thesis.researchQuestion`. State the question(s) precisely — well-formed, with appropriate scope, identifying what's empirical / theoretical / methodological.

4.3 Contribution

Pull from your `dataimago-spec.yaml`'s `vertical.dissertation.thesis.contribution`. State the contribution clearly. If you have multiple contributions, distinguish them (theoretical / methodological / empirical / practical) in a numbered list.

4.4 Roadmap

This dissertation proceeds as follows. Chapter 2 reviews the relevant literature. Chapter 3 details the methodology. Chapter 4 presents the results. Chapter 5 discusses implications. Chapter 6 concludes.

Chapter 5

Literature Review

5.1 Theoretical framework

Identify the theoretical traditions your work engages with. Cite foundational works using `[@bibtex-key]` syntax — entries live in `../references.bib`.

5.2 Current state of the field

What does the existing literature say about your research question? Map the landscape: who are the major voices, what positions do they hold, where are the gaps?

5.3 Positioning your contribution

Locate your work within the literature. What gap does it fill? What conversations does it extend? What does it dispute, if anything?

5.4 Synthesis

A short synthesis paragraph that motivates the methodology chapter.

Chapter 6

Methodology

6.1 Approach

Pull the high-level approach from your `dataimago-spec.yaml`'s `vertical.dissertation.thesis.method` field (quantitative / qualitative / mixed-methods / computational) and elaborate.

6.2 Data

Describe your data source(s). If you're using restricted data subject to a DUA, note that here + record details in `dataimago-spec.yaml`'s `vertical.dissertation.compliance.dua`.

6.3 Analysis

Describe the analytic techniques. If quantitative or computational, R code in the parent package's `R/` directory can be referenced here via embedded chunks:

```
```${r}
#| echo: true
Example: load the package + show its info
library(yourpkg)
package_info()
```

~ ~ ~

## 6.4 Ethical considerations

If your work required IRB approval, note the approval number here + record details in `dataimago-spec.yaml`'s `vertical.dissertation.compliance.irb`. The number typically appears on the dissertation's title page at many institutions.

## 6.5 Limitations

What are the methodological limits of this approach? Be honest. Committees appreciate candor about scope.



# Chapter 7

## Results

This chapter's structure depends entirely on your methodology + research question. Replace the placeholder sections below with sections appropriate to your work.

### 7.1 (Section 1)

(Replace this section.)

### 7.2 (Section 2)

(Replace this section.)

### 7.3 Summary of findings

(A short summary paragraph at the end of the results chapter — easy for your committee to skim. Lead with the most important findings.)

# **Chapter 8**

## **Discussion**

### **8.1 Interpretation**

What do the results mean? Connect them back to the research question stated in Chapter 1.

### **8.2 Theoretical implications**

How do your findings extend, challenge, or refine the theoretical frameworks identified in Chapter 2?

### **8.3 Practical / policy implications**

If your work has practical applications — for educators, clinicians, policy makers, practitioners — discuss them here.

### **8.4 Limitations + future work**

Be specific. What questions does your work leave open? What would the next study in this line look like?

# **Chapter 9**

## **Conclusion**

A short closing chapter (5–10 pages typical) restating the research question, summarizing the contribution, and offering a final reflection on the significance of the work.

### **9.1 Restating the research question**

Briefly remind the reader of the central question this dissertation answered.

### **9.2 Summary of contribution**

In two or three paragraphs, recap your contribution as the reader has now seen it argued + supported across the dissertation.

### **9.3 Closing reflection**

What did you learn in the process of doing this work? What surprised you? Where do you see your scholarly project going next?

## **Chapter 10**

### **Back matter**

# References

The bibliography is rendered automatically from `../references.bib`. No content needed here — Quarto handles it.

# Appendices

Optional. If you have appendices (data tables, supplementary analyses, code listings, interview protocols), add them as `chapters/99a-appendix-X.qmd`, `99b-appendix-Y.qmd`, etc. and reference them in `../_quarto.yml`.